

Anthropometry of the Nose from Adult Annangs and Ibibios of Akwa Ibom State, Nigeria

*Inoh MI,¹ Ekanem AU,² Inoh EE³ Bob B,²

ABSTRACT

Anthropometric parameters of the nose vary with age, sex, environment, and ethnic background. The nasal index is a standardised measurement used by Ear, Nose and Throat surgeons undertaking nasal reconstructions, anthropologists, and forensic experts to classify nose shapes. This study aimed to provide anthropometric data on the basic nasal parameters (nasal height, nasal width, and nasal index) of two ethnic groups (Ibibios and Annangs) in Akwa Ibom State, Nigeria. The study comprised of 600 subjects equally distributed between the two study groups, with their ages ranging from 18 to 60 years. The nasal height and nasal width were measured using a sliding Vernier caliper and the nasal index was calculated. Test for significance was done using an independent t-test. The results from the study showed that the mean nasal index for Ibibio males and females were 97.75 ± 15.63 and 93.65 ± 15.67 , respectively, while Annang males and females were 95.77 ± 12.85 and 93.46 ± 14.00 , respectively. Sexual dimorphism was noted in the nasal parameters of males and females in each ethnic group ($p < 0.05$). This shows that both ethnic groups have the platyrrhine nose type.

Keywords: Anthropometry, nasal width, nasal height, and nasal index.

INTRODUCTION

Human variation is a significant phenomenon that is influenced by a number of variables, including physical morphology, ecology, geographic location, racial and gender differences, mutations, and natural selection.^{1,2} Anthropometric measurements are crucial for understanding human population diversity.³ Though the importance of anthropometry to the study of various races was replaced in the 20th century by more advanced techniques for identifying racial variances, it has however recently become more widely used in the medical sciences, particularly in the fields of medical forensics and anthropology.⁴

Since the proportions of the nose are established on the same principle as the skull, it is widely recognized as one of the best indicators of racial origin.⁵ Both racial and environment influences have impact on the tip, the septum, and the nares of the nose.⁶ The study of nasal anthropometry involves

taking measurements of the human nasal proportions, size, and shape. This knowledge could be used in determining different races, ethnicity, and sex of an individual.⁷ Additionally, nasal anthropometric proportions and indices have been employed in rhinoplasty, cosmetic and plastic reconstructive surgery, as well as criteria for measurement of facial attractiveness.^{8,9}

Parameters such as height, width, and nasal index can be used to categorize the nose. The nasal height is the distance between the nasion and sub-nasal septum (the point where the nasal septum touches the upper lip), while the nasal width is the widest distance between the two alae nasi.¹⁰ The nasal index is the ratio of the nasal width to the nasal height multiplied by 100, and is also referred to as the percentage ratio of the breadth and height of the nose.^{11,12} It is a standardized measurement used by Ear, Nose and Throat surgeons undertaking nasal and facial reconstructions, anthropologists, and forensic experts to classify nose shapes.^{7,9} Anthropologists apply the nasal index in the categorization of many ethnic groups, gender discrimination, and the classification of the human race.¹³⁻¹⁶ The nasal index is also useful in dysmorphology, in the early diagnosis of some dysmorphic syndromes like cleft lip and cleft palate, which are associated with nose disorders.⁷

Department of Surgery, ¹Faculty of Clinical Sciences, University of Uyo.

Department of Human Anatomy, ²Faculty of Basic Medical Sciences, University of Uyo.

Department of Community Medicine, ³Faculty of Clinical Sciences, University of Uyo.

*Corresponding Author: dr.inohmi@gmail.com

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Based on nasal parameters, the human nose can be classified into leptorrhine (long and narrow nose), mesorrhine (medium nose), and platyrrhine (broad nose) types. The leptorrhine nose type has a nasal index of $\leq 69.9\%$, the mesorrhine nose type has a nasal index of between 70 – 84.9%, while the platyrrhine nose type has a nasal index of $\geq 85\%$.^{1,13,17} Leptorrhine nose types have the least prominent alar lobule with a well-defined nasal tip. The mesorrhine nose has a less prominent lobule with a more defined nasal tip, while the platyrrhine nose is characterized by a very prominent alar lobule with a full or well-rounded nasal tip.¹³

In Africa, particularly in south-south Nigeria, beauty is a tool used for both tribal and marital recognition. The human face and in particular the nose, play important roles in the physiognomic features of humans.^{18,19} There are several studies concerning nasal indices among ethnic groups in Nigeria, but this is the first amongst the Annangs and Ibibios. Data are still lacking on whether individuals from different ethnic groups like the Annang and Ibibio that share the same geography have the same nose type since climate and environment are believed to influence the variety of noses.²⁰ This study aimed to document the baseline data of nasal height, width, and nasal index for adult males and females from the Annang and Ibibio ethnic groups of South-south, Akwa Ibom state, Nigeria.

MATERIALS AND METHODS

Study Population

The study was carried out using a total sample size of 600 subjects randomly drawn from two ethnic groups in Akwa Ibom State, which comprised 300 from each of the two ethnic groups of the Annangs and Ibibios. The subjects comprised individuals with normal craniofacial configuration, between the ages of 18 to 60 years. The subjects were

natives of Ibibio and Annang ethnicity, resident in their indigenous communities, and both parents having their origin from the same tribe. Subjects with a previous history of trauma to the nose and/or face, history of rhinoplasty, facial surgeries, sinonasal tumours, facial dysmorphism, and congenital abnormalities, were excluded from the study.

Ethical Approval

Ethical approval was obtained from the Faculty of Basic Medical Science Research and Ethical Committee, University of Uyo. The consent of the subjects was also sought before the collection of data.

Measurement of Nasal Parameters

The direct method described by Anas and Salah was used.²⁰ Accordingly, measurements were taken with the subject sitting on the chair in an upright and relaxed condition and the head in an anatomical position. The measurements were undertaken by one observer to prevent inter-observer error and each measurement was taken twice and the average was recorded to reduce the error of measurements. It was ensured that the caliper was placed properly and accurate reading was taken. Each measurement was taken to the nearest 0.01mm. It was ensured that each subject did not smile or change facial expressions while taking measurements in order to get accurate values.

The digital sliding Vernier caliper after being adjusted to an accurate point was used in measuring the Nasal Height (NH): measured from the nasion to the sub-nasal septum; Nasal Width (NW): maximum width of the nose, measured at a right angle to the nasal height from right ala to left ala (Figure 1). The nasal index was calculated as follows: Nasal index = [Nasal Width (NW) / Nasal Height (NH)] multiplied by 100.²⁰ Mathematically:

$$\text{Nasal Index (NI)} = \frac{\text{Nasal width}}{\text{Nasal height}} \times 100$$

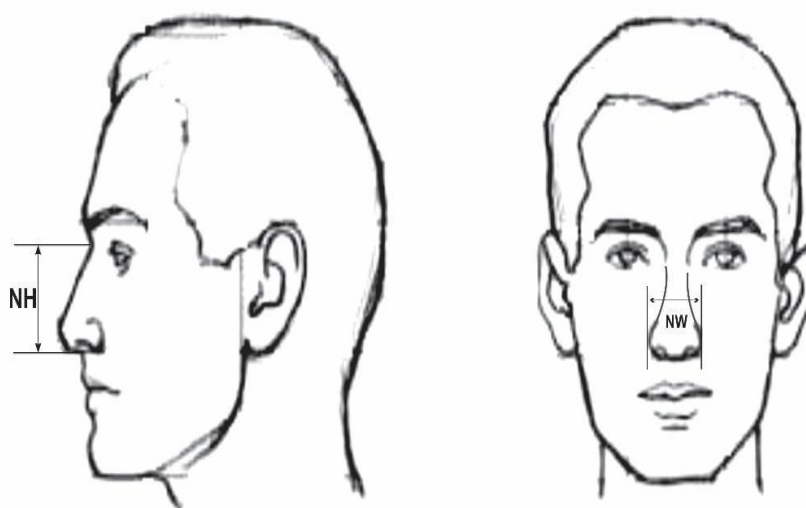


Figure 1: Measurement of the Nasal height (NH) and nasal width (NW).

[Adapted from - br.pinterest.com/pin/482377810089823374]

Limitations of the study

Facial measurements (height and width) were not taken into consideration in this study, but this can be incorporated into subsequent studies.

Statistical Analysis

Data were analyzed using SPSS Windows version 25.0. Descriptive statistics of mean $\pm$ SD and percentage were used to summarize the data obtained. Two-sided values were calculated using the paired samples T-test for observing variables. p values < 0.05 were considered statistically significant.

RESULTS

The results consisting of the statistical analysis with respect to the nasal parameters (nasal height, nasal width, and nasal index) of the two ethnic groups are shown in table 1. The mean nasal width among the Ibibio males was 4.25 ± 0.49 cm, while the females had 4.19 ± 0.50 cm, with a very significant difference ($p < 0.01$) between the sexes. Among the Annangs the mean nasal width was 4.23 ± 0.39 cm for the males, while the females had a mean value of 4.13 ± 0.50 cm. The mean values also indicated significant

($p < 0.01$) sexual dimorphism among the Annangs.

The mean nasal height was 4.41 ± 0.53 cm and 4.30 ± 0.55 cm among the Ibibio males and females, respectively; while the mean nasal height of the Annangs was 4.44 ± 0.50 cm for the males and 4.13 ± 0.50 cm for the females. The mean nasal height in both ethnic groups showed significant ($p < 0.01$) sexual dimorphism.

The nasal indexes calculated from the nasal parameters were about 97.75 and 93.65 for the males and females of Ibibio origin, respectively. Among the Annangs the males had a nasal index of 95.77 and the females had 93.46. The males had significantly ($p < 0.05$) higher values compared to the females.

Analysis of the results based on the two ethnic groups gave the mean nasal height of the Ibibios to be 4.41 ± 0.54 cm and that of the Annangs was 4.41 ± 0.50 cm. The combined mean nasal width was 4.22 ± 0.49 cm for the Ibibios and 4.13 ± 0.49 cm for the Annangs. The combined male and female nasal indices of 97.05 and 94.61 were obtained respectively, for the Ibibios and Annangs. The results showed that all the subjects had platyrrhine nose type.

Table 1: The Mean Nasal Width, Nasal Height, and Nasal Index of the Ibibio and Annang Ethnic Groups Based on Gender

Group	Ibibio		Annang	
	Males (N=150)	Females (N=150)	Males (N=150)	Females (N=150)
Nasal Width (Mean±SD cm)	4.25±0.49	4.19±0.50**	4.23±0.39	4.13±0.50**
Nasal Height (Mean±SD cm)	4.41±0.53	4.30±0.55**	4.44±0.50	4.37±0.50**
Nasal Index	97.75±15.63	93.65±15.67*	95.77±12.85	93.46±14.00*

Key: SD - Standard Deviation; N-Number of Subjects; Significant at **p<0.01; *p<0.05

Table 2: Mean Nasal Parameters of the Ibibio and Annang Ethnic Groups

Group	Nasal Height (cm)		Nasal Width (cm)		Nasal Index	
	Ibibio	Annang	Ibibio	Annang	Ibibio	Annang
Mean±SD	4.41±0.54	4.41±0.50	4.22±0.49	4.13±0.49	97.05±15.63	94.61±13.44
N	300	300	300	300	300	300

Key: SD - Standard Deviation, N - Number of Subjects

DISCUSSION

The morphological difference in nasal anatomy cannot be overemphasized, especially in rhinoplasty and reconstructive facial plastic surgeries. The shape of the nose is believed to be a unique feature of the ethnic origin of an individual; as such, the nasal anthropometric features, when studied, can provide useful information to the ear, nose and throat surgeon especially when nasal reconstruction is envisaged.²¹ Natural selection in human evolution could cause certain variables like race, tribe, and environmental climatic conditions to determine the shape of the nose, with the narrower nose being favored in cold weather and dry climate, while the flat or wide nose could be seen mostly in people living in areas that are moist and warm.^{6,22,23} Anthropometric parameters of the nose also vary with age, sex, and ethnic background.²⁴

The use of anthropometry to establish nasal dimensions has been reported by many authors. In Nigeria, Akpa *et al.* and Oladipo *et al.* respectively did a morphometric study on nasal parameters among Nigerians of Igbo descent, in south-eastern Nigeria.^{19,25} Akpa

et al. classified the nose type of the Igbos under the platyrrhine category.²⁵ Oladipo *et al.* studied the nasal parameters of Igbos, Ijaw, and Yoruba ethnic groups in southern Nigeria.¹⁹ Their findings showed a mean nasal index of > 85.0 in the three Nigerian ethnic groups studied, and that on average, the Igbo had a mean nasal index of 94.1 ± 0.37 , Yorubas 89.2 ± 0.3 and the Ijaws 96.37 ± 1.06 ; all of which are the platyrrhine nose type. A study in south-south Nigeria on anthropometric assessment of nasal parameters of the Itsekiri and Urhobo ethnic groups of Nigeria showed that on average, the Urhobos have a mean nasal index of 89.63% while the Itshekiris have 90.74%.¹⁰ Therefore, these works show that the two ethnic groups fall within the same nose type which is platyrrhine (broad and short) expected from an African population. Sexual dimorphism was observed in the 2 ethnic groups studied, with the males having significantly ($p<0.05$) higher nasal index than the females.

In the present study, the nasal index of the Ibibio and Annang ethnic groups (male and female) places them under the platyrrhine

nose type. This partly confirms the previous study carried out by Oladipo and colleagues, which reported a platyrrhine nose type in an assessment of the nasal indices of Igbo, Ijaw, and Yoruba ethnic groups in Southern Nigeria.¹⁹ In another study carried out among the Ibibio and Yakurr ethnic groups in South-south Nigeria; Eliakim and colleagues also reported a platyrrhine nose type.²⁶ However, a few studies have shown that mesorrhine nose type exists in some ethnic groups of Nigeria. Oladipo identifies a mesorrhine nose type for the Adoni ethnic group of Rivers State in Nigeria.²⁷ This reveals that a racial variation still exists in the nasal index of various populations. The reason for the platyrrhine nose type can be linked to the assertion made by Hall and Hall that a broader nose or platyrrhine nose type is favored in hot, humid climates.¹¹ This distribution is identified as a consequence of natural selection.

The Ibibio and Annang adult males have significantly higher nasal height and nasal width compared to their female counterparts, indicating sexual dimorphism. Eliakim *et al.* reported higher values of nasal

parameters among the Yakurr females in the study of nasal parameters of Ibibio and Yakurr ethnic groups of south-south Nigeria.²⁶ The reason for sexual dimorphism may be due to biological factors such as genetics and hormones.²⁸ In our study, the ethnic populations (Ibibios and Annangs), though live in similar geographical regions, they are in separate locations with their distinct socio-cultural values. The variation observed in our study is more related to genetic variability than the environment.

CONCLUSION

In conclusion, the adult nose of Ibibio and Annang descent of Nigeria both fall under the platyrrhine nose type based on the nasal index. This study also shows that there is the existence of sexual dimorphism in all parameters considered in each ethnic group with the males having significantly higher values than the females.

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